Kapture Finance Collections Voicebot

High-Level Design (HLD) — AI Delivery Intern Take-Home Assignment

Agent: Maya   **Channel:** Outbound Voice   **Platform target:** Vapi

Purpose

Design an outbound collections voice agent that politely contacts customers with overdue loan EMIs, authenticates the customer before disclosing debt information, understands intent, captures a payment commitment when appropriate, and escalates cases that require human support.

Reference scenario: Rahul Sharma • Personal loan • Overdue EMI ■8,499 • 12 days past due.

Design principles

1. Verification before disclosure. 2. State-enforced controls rather than prompt-only discretion. 3. Clear, respectful collections language. 4. Tool calls for authoritative account actions. 5. Every call ends with a disposition. 6. Safe handling of wrong-person, disputes, hardship, opt-out and hostile interactions.

1. Architecture & Pipeline

The proposed pipeline follows the assignment's required pattern: telephony → STT → orchestrator/LLM → TTS, with business tools and a datastore behind the orchestrator.

Customer	Telephony / Vapi	STT	Orchestrator + LLM	TTS	Customer
Answers outbound call	Call session, routing	Speech → text	State machine, policy, intent, tool calls	Text → speech	Hears Maya

Supporting services: account datastore/CRM, customer verification service, payment-link service, disposition store, escalation/handoff endpoint, and observability/logging.

Latency budget (target): telephony/Vapi overhead 150–300 ms; STT partial/final response 300–700 ms; orchestrator/LLM decision 500–1,000 ms; tool API 100–500 ms; TTS first audio 300–700 ms. Target perceived turn latency: roughly 1.5–2.5 seconds for normal turns, with streaming used where available.

Data boundary: the LLM should receive only the minimum fields needed for the current state. Account balance, overdue amount and payment status come from tools rather than model memory.

2. Conversation Flow / State Machine

State	Purpose / allowed action	Transition / lock
INIT	Start call; identify company/agent; no debt disclosure.	→ AUTH_PENDING after customer engages.
AUTH_PENDING	Verify identity using approved non-sensitive factors. No overdue amount or loan details.	Locked: debt disclosure tools unavailable until AUTH_VERIFIED.
AUTH_VERIFIED	Disclose purpose and overdue amount; ask intent.	→ WILL_PAY / HARDSHIP / DISPUTE / ALREADY_PAID / CALLBACK / DNC / ESCALATE.
WILL_PAY	Agree amount/date; validate PTP fields.	→ PTP_CAPTURED after valid date + amount.
HARDSHIP	Acknowledge difficulty; do not pressure; offer escalation/options supported by policy.	→ ESCALATE or CALLBACK.
DISPUTE	Do not argue; capture dispute reason and escalate.	→ ESCALATE.
ALREADY_PAID	Acknowledge; avoid demanding payment; verify/log claimed payment and route for review.	→ ESCALATE or CLOSE.
WRONG_PERSON	Do not disclose debt; politely end and disposition as wrong person.	→ CLOSE.
CALLBACK	Capture requested callback time if supported.	→ CALLBACK_LOGGED → CLOSE.
DNC	Stop collections conversation and log opt-out.	→ CLOSE.
ESCALATE	Transfer or create human-agent case with context.	→ HUMAN_HANDOFF / CLOSE.
CLOSE	Confirm next step and disposition; end call.	Terminal state.

3. Intents & Entities

Intent	Entities / fields	Action
will-pay	PTP date, PTP amount, payment method (if allowed)	Confirm commitment; log PTP.
cannot-pay / hardship	Reason, affordability issue, requested assistance	Empathize; offer approved options or escalate.
dispute	Dispute reason, amount questioned	Do not argue; escalate/review.
already-paid	Payment date, reference if volunteered	Do not demand payment; log claim and route.
wrong-person	Relationship / refusal to identify	No debt disclosure; close.
callback-request	Preferred date/time, timezone	Log callback if supported.
hostile	None required	De-escalate once; if continuing, end/escalate.
do-not-call	Opt-out request	Stop collections conversation and log DNC.
no-input / voicemail	None	Retry/leave only permitted message; otherwise disposition.

4. Tools / API Contracts

Tool	Inputs	Outputs / purpose
get_account_details	customer_id / internal call context	account status, product, overdue amount, days past due; only callable after verified state.
verify_customer	customer_id, approved verification factors	verified=true/false; reason/code. Does not return debt data on failure.
log_promise_to_pay	customer_id, amount, date, call_id	success, PTP id; writes commitment to datastore.
send_payment_link	customer_id, channel, call_id	success, link/reference; never expose raw sensitive data.
escalate_to_agent	call_id, reason, summary, customer_id	handoff/ticket id.
mark_disposition	call_id, disposition, metadata	success; terminal call outcome recorded.

5. Authentication & Data Safety

Authentication is a hard state gate. The assistant begins in AUTH_PENDING and cannot call debt-disclosure/account-detail operations until verify_customer returns verified=true. If another person answers, the bot must not reveal that the call concerns a loan, overdue EMI, amount, lender account or collections.

Recommended pattern: ask for approved verification factors that are safe to request from the person answering; send only the minimum required values to verify_customer; retain verification result against call_id/session state; expire verification when the call ends.

The datastore should separate call metadata, verification result, account data and dispositions. Logs should avoid unnecessary sensitive information. Tool responses should be structured and authoritative; the LLM should not invent balances, dates, fees, policies or payment outcomes.

6. Guardrails & Compliance

• Mandatory self/company/purpose disclosure before collection discussion.
• Do not disclose debt to an unverified person or wrong number.
• Use only permitted calling hours and approved contact policy.
• No threats, harassment, intimidation, misleading claims or pressure beyond approved collection policy.
• Honor do-not-call/opt-out requests immediately and record the disposition.
• Never fabricate account status, payment receipt, settlement terms, fees, deadlines or legal consequences.
• Stay on collections scope; refuse unrelated requests and return to the call objective.
• Use calm, respectful language; if the caller becomes abusive, de-escalate and end/escalate safely.
• Use human escalation when the customer disputes, requests unsupported exceptions, shows hardship requiring review, or when the agent cannot safely proceed.

7. Edge Cases

Case	Expected behavior
Already paid	Acknowledge claim; do not continue payment pressure; capture available reference/date and route for review.
Dispute amount	Do not argue. Capture reason and escalate to a human/review queue.
Do not call	Immediately stop collection conversation, log DNC, close politely.
Wrong number/person	No debt disclosure. Confirm only what is necessary and end/disposition.
Voicemail/no input	No sensitive debt disclosure. Follow approved voicemail policy; retry or close.
Hostile caller	One calm de-escalation attempt; if hostility continues, end or escalate.
Language switch EN/HI	Switch language cleanly if supported; preserve the same state and guardrails.
Tool/API failure	Do not guess. Explain briefly, retry within safe limits or escalate; log technical failure.
Customer asks unrelated question	Acknowledge and redirect to the collections purpose; do not expose unrelated internal data.

8. Escalation & Disposition

A human handoff should occur for disputes, complex hardship, unsupported requests, repeated verification failure, policy-sensitive situations, or explicit requests for a human. The handoff payload should contain call_id, verified status, intent, concise summary, relevant tool results and the requested next action.

Every call must finish with one terminal disposition, for example: PTP_CAPTURED, HARDship_ESCALATED, DISPUTE_ESCALATED, ALREADY_PAID_REVIEW, CALLBACK_REQUESTED, DNC, WRONG_PERSON, NO_ANSWER, VOICEMAIL, HOSTILE_ENDED, TECHNICAL_FAILURE, or HUMAN_HANDOFF.

9. Observability

Metric	Why track it
Containment rate	How many routine calls complete without human intervention.
PTP rate	Measures successful payment commitments among eligible conversations.
PTP kept rate	Measures downstream quality of commitments, not just promises.
Average turn latency	Finds STT/LLM/tool/TTS bottlenecks.
Call drop / failure rate	Detects telephony and platform reliability issues.
Verification success rate	Detects authentication friction or data issues.
Escalation rate by intent	Shows which cases need better automation or policy support.
DNC rate	Monitors opt-out handling and customer contact patterns.
Tool error rate	Detects backend/webhook failures.

10. End-to-End Example

Scenario: Rahul Sharma answers Maya's outbound call. Maya introduces herself and Kapture Finance, explains that the call concerns an account only after successful verification, and states the overdue EMI of ■8,499. Rahul says he can pay on a specified date. Maya confirms the proposed amount/date, calls log_promise_to_pay, optionally offers the approved payment-link flow, confirms the next step and calls mark_disposition with PTP_CAPTURED.

Edge path: If Rahul says the EMI was already paid, Maya does not argue or demand another payment. She acknowledges the statement, captures the payment date/reference if voluntarily provided, routes the case for review, and ends with ALREADY_PAID_REVIEW.

11. Implementation Notes / Assumptions

This HLD is intentionally platform-oriented rather than production-specific. Vapi is the target voice platform from the assignment. Backend endpoints may be mocked for the take-home build, provided their contracts and state restrictions are demonstrated. The exact STT/TTS/model vendors are implementation choices and should be documented in Task 2 with the rationale for the selections.

The key engineering requirement is that authentication and disclosure rules are enforced by application state/tool permissions, not merely by instructions in the LLM prompt.

Prepared for: Kapture AI Delivery Intern Take-Home Assignment